Family name:_____ Given name: _____

Exam

Advanced Topics in Engineering Physics ('Plasmas')

Constants (approximated values):

- $e = 1.60 \cdot 10^{-19}$ A·s
- $h = 6.63 \cdot 10^{-34}$ Js
- $k_B = 1.38 \cdot 10^{-23}$ m²·kg·s^{-2}·K^{-1}
- $m_e = 9.11 \cdot 10^{-31}$ kg
- $\varepsilon_0 = 8.85 \cdot 10^{-12}$ A·s·V^{-1}·m^{-1}
- $c = 3 \cdot 10^8$ m/s
- 1 eV = $1.6 \cdot 10^{-19}$ J

1.
Please indicate whether the following statements are true or false by checking the appropriate box (Note: Only one box can be checked and only right answers are considered whereas wrong ones are disregarded, so there are no 'negative points' possible.)

Statement	True	False
Plasmas are always fully ionised.		
The dissociation energy of a gas is notably lower than its ionisation energy.		
Plasmas are quasi neutral; there is not charge on a macroscopic scale.		
The term 'Debye length' is another expression for the length of a plasma.		
In an equilibrium plasma, the electron temperature is lower than both the ion temperature and neutral gas temperature.		
The Townsend discharge is a self-maintaining discharge.		

2.
A plasma source consist of two plane-parallel circular metallic electrodes. The diameter of the electrodes is D = 26 mm and the distance (= air gap thickness; resistor of air $R = 1.5 \cdot 10^{14}$ Ω) is d = 5 cm. The voltage applied to the electrodes is V = 150 V.

a. Determine the electric field E between the electrodes.
b. Determine the current density J between the electrodes.
c. Determine the drift velocity v_{Drift} of charged particles within this plasma where the particle's mobility amounts to 74 A·s²/kg.

3.
Determine the

a. electron density n_e and
b. the plasma frequency ω_p

of a plasma in thermodynamic equilibrium where the following values are known:

- Plasma temperature T = 11604K
- Debye radius λ_D = 235 µm.

4.
A plasma is operated at atmospheric pressure; the used process gas is molecular nitrogen with an ionisation energy of E_{ion} = 15.7 eV. Determine the 1^{rst} Townsend coefficient α for electrons with an energy of $E_{e-} = 2 \cdot 10^{-19}$ J.

EXAMPLE

- Please turn over -

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5.
A plasma is ignited using argon as process gas (ionisation energy E_{ion} = 15.75 eV, mass m_{Ar} = $6.63 \cdot 10^{-26}$ kg). Within this plasma, an electron is generated via photoelectric emission at a copper electrode surface. The work function of this surface is W = 4.55 eV and the wavelength of the incident light is 70 nm.

a) Does the generated electron provide sufficient kinetic energy for the ionisation of argon (neglecting a possible acceleration of the electron by the electric field)?
b) Determine both the maximum energy E (given in eV) and temperature T (given in K) of argon ions within the plasma assuming an electron energy as determined in a). (hint: 1 eV = 11604K)
c) Determine the potential (given in eV) within the sheath of this plasma.

6.
An atmospheric pressure plasma in perfect local thermal equilibrium is operated at a voltage of V = 17 kV where the discharge gap thickness is d = 2 mm. Determine the energy of electrons within this plasma.

- Good luck! -

EXAMPLE